

TASK-1

The image shows a Wireshark packet capture of network traffic. The top pane displays a list of packets. The middle pane shows the details of the selected packet (No. 19716), which is a QUIC packet. The bottom pane shows the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
19698	101.652298080	2409:40c1:4149:231b::	2404:6800:4002:829::	QUIC	97	Protected Payload (KP0), DCID=fdf65e23c01bc7e
19699	101.707674700	2404:6800:4002:829::	2409:40c1:4149:231b::	QUIC	86	Protected Payload (KP0)
19700	102.375150600	2409:40c1:4149:231b::	2405:200:1609:1711::	TCP	74	ACK Retransmission [54537 + 443] [FIN, ACK] Seq=1 Ack=1 Win=1019 Len=0
19701	102.442421500	2409:40c1:4149:231b::	2606:4700:8d72:7cbb::	TCP	1374	58001 + 443 [ACK] Seq=3779787 Ack=8013456 Win=2096128 Len=1300 [TCP PDU reassembled in 19707]
19702	102.442421500	2409:40c1:4149:231b::	2606:4700:8d72:7cbb::	TCP	1374	58001 + 443 [ACK] Seq=3781087 Ack=8013456 Win=2096128 Len=1300 [TCP PDU reassembled in 19707]
19703	102.442421500	2409:40c1:4149:231b::	2606:4700:8d72:7cbb::	TCP	1374	58001 + 443 [ACK] Seq=3782387 Ack=8013456 Win=2096128 Len=1300 [TCP PDU reassembled in 19707]
19704	102.442421500	2409:40c1:4149:231b::	2606:4700:8d72:7cbb::	TCP	1374	58001 + 443 [ACK] Seq=3783687 Ack=8013456 Win=2096128 Len=1300 [TCP PDU reassembled in 19707]
19705	102.442421500	2409:40c1:4149:231b::	2606:4700:8d72:7cbb::	TCP	1374	58001 + 443 [ACK] Seq=3784987 Ack=8013456 Win=2096128 Len=1300 [TCP PDU reassembled in 19707]
19706	102.442421500	2409:40c1:4149:231b::	2606:4700:8d72:7cbb::	TCP	1374	58001 + 443 [ACK] Seq=3786287 Ack=8013456 Win=2096128 Len=1300 [TCP PDU reassembled in 19707]
19707	102.442421500	2409:40c1:4149:231b::	2606:4700:8d72:7cbb::	TLSv1.3	404	Application Data
19708	102.442520700	2409:40c1:4149:231b::	2606:4700:8d72:7cbb::	TCP	1374	58001 + 443 [ACK] Seq=3787917 Ack=8013456 Win=2096128 Len=1300 [TCP PDU reassembled in 19711]
19709	102.442520700	2409:40c1:4149:231b::	2606:4700:8d72:7cbb::	TCP	1374	58001 + 443 [ACK] Seq=3789217 Ack=8013456 Win=2096128 Len=1300 [TCP PDU reassembled in 19711]
19710	102.442520700	2409:40c1:4149:231b::	2606:4700:8d72:7cbb::	TCP	1374	58001 + 443 [ACK] Seq=3790517 Ack=8013456 Win=2096128 Len=1300 [TCP PDU reassembled in 19711]
19711	102.442520700	2409:40c1:4149:231b::	2606:4700:8d72:7cbb::	TLSv1.3	422	Application Data
19712	102.472184600	2606:4700:8d72:7cbb::	2409:40c1:4149:231b::	TCP	74	443 + 58001 [ACK] Seq=8013456 Ack=3782387 Win=7593984 Len=0
19713	102.472268100	2606:4700:8d72:7cbb::	2409:40c1:4149:231b::	TCP	86	443 + 58001 [ACK] Seq=8013456 Ack=3783687 Win=7602176 Len=0 SLE=3787587 SRE=3787917
19714	102.472309800	2606:4700:8d72:7cbb::	2409:40c1:4149:231b::	TCP	74	443 + 58001 [ACK] Seq=8013456 Ack=3787917 Win=7610368 Len=0
19715	102.472310300	2606:4700:8d72:7cbb::	2409:40c1:4149:231b::	TCP	74	443 + 58001 [ACK] Seq=8013456 Ack=3789217 Win=7610368 Len=0
19716	102.472310600	2606:4700:8d72:7cbb::	2409:40c1:4149:231b::	TCP	74	443 + 58001 [ACK] Seq=8013456 Ack=3792165 Win=7610560 Len=0

Frame 1: Packet, 75 bytes on wire (600 bits), 75 bytes captured (600 bits) on interface \Device\NPF_{A0F...} Ethernet II, Src: f6:db:87:20:0b:1d (f6:db:87:20:0b:1d), Dst: ba:a7:47:f0:6a:41 (ba:a7:47:f0:6a:41) Internet Protocol Version 6, Src: 2409:40c1:4149:231b::f0c0:b984:8e12:5db0, Dst: 64:ff9b:136f4:651f 0110 = Version: 6 0000 0000 = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT) 0000 00 = Differentiated Services Codepoint: Default (0) 0110 1100 0111 1010 0111 = Flow Label: 0x6c7a7 Payload Length: 21 Next Header: TCP (6) Hop Limit: 63 Source Address: 2409:40c1:4149:231b::f0c0:b984:8e12:5db0 Destination Address: 64:ff9b:136f4:651f [Embedded IPv4 Prefix: 0064ff9b0000000000000000] [Destination Embedded IPv4: 54.244.101.31] [Embedded IPv4: 54.244.101.31] [Stream index: 0] Transmission Control Protocol, Src Port: 64222, Dst Port: 443, Seq: 1, Ack: 1, Len: 1

Protocols observed: 1. QUIC 2. TCP (Transmission Control Protocol) 3. TLSv1.3 (Transport Layer Security) 4. IPv6 (Internet Protocol Version 6) 5. Ethernet II

Description of the protocols:

- QUIC: A modern transport protocol used by HTTP/3 to provide secure and faster web communication.
- TCP: A reliable, connection-oriented protocol used for transmitting data between devices.
- TLSv1.3: Encrypts data exchanged between the web browser and the server to ensure secure communication.
- IPv6: The network-layer protocol used to route packets across the Internet.

TASK-2

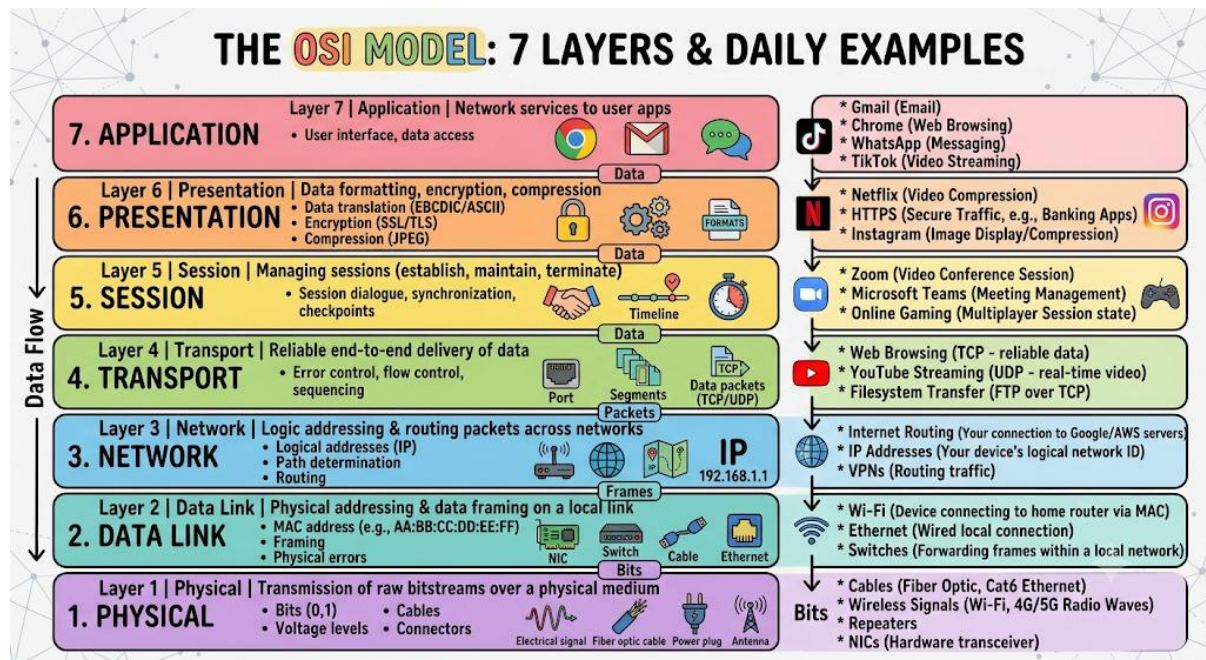
The image shows a Wireshark packet capture of network traffic. The top pane displays a list of packets. The middle pane shows the details of the selected packet (No. 1), which is an HTTP packet. The bottom pane shows the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	4.240.189.68	10.31.192.192	HTTP	1334	Continuation
2	0.003979000	4.240.189.68	10.31.192.192	TCP	66	80 + 52226 [ACK] Seq=1 Ack=1 Win=22 Len=0 TSval=2001331620 TSecr=983863
3	0.004551700	4.240.189.68	10.31.192.192	HTTP	1334	Continuation
4	0.004607500	10.31.192.192	4.240.189.68	TCP	66	52227 + 80 [ACK] Seq=1 Ack=2537 Win=319 Len=0 TSval=983891 TSecr=2001331616
5	0.004681400	4.240.189.68	10.31.192.192	HTTP	322	Continuation
6	0.004738300	4.240.189.68	10.31.192.192	HTTP	1334	Continuation
7	0.004761000	10.31.192.192	4.240.189.68	TCP	66	52227 + 80 [ACK] Seq=1 Ack=4061 Win=319 Len=0 TSval=983891 TSecr=2001331621
8	0.007137800	4.240.189.68	10.31.192.192	HTTP	1334	Continuation
9	0.007255100	4.240.189.68	10.31.192.192	HTTP	1334	Continuation
10	0.007294400	10.31.192.192	4.240.189.68	TCP	66	52227 + 80 [ACK] Seq=1 Ack=6597 Win=319 Len=0 TSval=983894 TSecr=2001331621
11	0.007342200	4.240.189.68	10.31.192.192	HTTP	1334	Continuation
12	0.007343100	4.240.189.68	10.31.192.192	HTTP	1334	Continuation
13	0.007343900	4.240.189.68	10.31.192.192	HTTP	1334	Continuation
14	0.007344600	4.240.189.68	10.31.192.192	HTTP	1334	Continuation
15	0.007345300	4.240.189.68	10.31.192.192	HTTP	1334	Continuation
16	0.007399100	10.31.192.192	4.240.189.68	TCP	66	52227 + 80 [ACK] Seq=1 Ack=12937 Win=319 Len=0 TSval=983894 TSecr=2001331621
17	0.007441700	4.240.189.68	10.31.192.192	HTTP	1334	Continuation
18	0.007471600	10.31.192.192	4.240.189.68	TCP	66	52227 + 80 [ACK] Seq=1 Ack=14205 Win=319 Len=0 TSval=983894 TSecr=2001331621
19	0.007645000	4.240.189.68	10.31.192.192	HTTP	1334	Continuation

Frame 1: Packet, 1334 bytes on wire (10672 bits), 1334 bytes captured (10672 bits) on interface \Device\NPF_{A0F...} Ethernet II, Src: 62:34:d1:af:80:a6 (62:34:d1:af:80:a6), Dst: ce:32:ba:fa:0d:f7 (ce:32:ba:fa:0d:f7) Internet Protocol Version 4, Src: 4.240.189.68, Dst: 10.31.192.192 Transmission Control Protocol, Src Port: 80, Dst Port: 52227, Seq: 1, Ack: 1, Len: 1268

Packet No. Packet 1 Packet 2 Protocol OSI Layer HTTP Layer 7 – Application TCP Layer 4 – Transport Packet 1 (Packet Details) IPv4 Layer 3 – Network Brief explanation: • HTTP – Application Layer (Layer 7): Used to transfer web pages and web content. • TCP – Transport Layer (Layer 4): Provides reliable communication between devices. • IPv4 – Network Layer (Layer 3): Responsible for addressing and routing packets across networks.

TASK-3



TASK-4

How a WhatsApp Message Travels Through the OSI Model

When I type "Hello!" in WhatsApp and press **Send**, the message passes through several OSI layers before it reaches the other person's phone.

1. Application Layer (Layer 7)

This is where WhatsApp works. I type my message, and the app prepares it to be sent over the internet.

2. Presentation Layer (Layer 6)

The message is converted into a format that both devices can understand. WhatsApp also encrypts the message so that only the sender and receiver can read it.

3. **Session Layer (Layer 5)**

This layer manages the connection between my phone and WhatsApp's servers. It keeps the communication active while the message is being sent and received.

4. **Transport Layer (Layer 4)**

The message is divided into small pieces called **segments**. This layer ensures the data is delivered correctly and, if any part is lost, it is sent again.

5. **Network Layer (Layer 3)**

The message is given IP addresses and routed through the internet. This layer finds the best path from my phone to the receiver's device.

6. **Data Link Layer (Layer 2)**

The data is packed into **frames** and sent to my Wi-Fi router or mobile network using the device's MAC address. It also checks for transmission errors on the local network.

7. **Physical Layer (Layer 1)**

Finally, the data is transmitted as electrical signals, radio waves (Wi-Fi/mobile data), or light signals through cables and network equipment. The receiver's phone receives these signals, and the data moves back up the OSI layers until the WhatsApp message appears on the screen.